

Prevalence of Yersinia enterocolitica in market weight hogs in the United States.

Pigs are the major animal reservoir for *Yersinia enterocolitica* strains, which are potentially pathogenic for humans. The goals of this study were (i) to estimate the individual animal and on-farm prevalences of *Y. enterocolitica* in hogs based on tonsil samples collected during National Animal Health Monitoring System Swine 2002 study and (ii) to use these data with data previously published for fecal samples to determine on-farm risk factors for *Y. enterocolitica*. Tonsil swabs (1,218) and fecal samples (2,847) were collected on 124 farms located in the top 17 pork-producing states. Ten percent of tonsils (122 of 1,218 samples) were positive in irgasan-tiracillin-chlorate (ITC) enrichment broth by real-time PCR, but only 5.6% of samples (68 of 1,218) were positive after subculture on the more selective cefsulodin-irgasan-novobiocin (CIN) agar. For tonsils, the on-farm prevalence based on real-time PCR detection of the *ail* gene in ITC enrichment broth cultures was 32% (32 of 100 premises sampled); the prevalence based on subculture in CIN agar was 19.6% (20 of 102 premises). Results of bacteriological isolation and real-time PCR analysis of tonsils and feces were combined to estimate prevalence (individual animal and farm), which was subsequently correlated with 40 farm management practices. Four factors and their accompanying odds ratios (ORs) were identified in the final regression model: location in a central state (OR = 0.3), vaccination for *Escherichia coli* (OR = 3.0), percentage of deaths due to scours (OR = 3.5), and presence of meat or bone meal in grower-finisher diet (OR = 4.1).